

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/cage14.mtx 8 1

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 1

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/nlpkkt120.mtx 8 1

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/parabolic_fem.mtx 8 1

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/RM07R.mtx 8 1

Sigma = 1

```
ptfs214h@f0757:~/SpMVMS$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/cage14.mtx 8 1
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 1505785x1505785
  number of nonzeros: 27130349 (18.0174 per Row)
Starting CSR
runtime CSR: 0.487573 sec. performance: 1.11287e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 0.503765 sec. performance: 1.0771e+10 Flops/s, overhead: 2.73208%
```

```
ptfs214h@f0757:~/SpMVMS$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 1
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 1504002x1504002
  number of nonzeros: 110879972 (73.7233 per Row)
Starting CSR
runtime CSR: 1.65407 sec. performance: 1.34069e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 1.65339 sec. performance: 1.34124e+10 Flops/s, overhead: 0.274206%
```

```
ptfs214h@f0757:~/SpMVMS$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/nlpkkt120.mtx 8 1
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 3542400x3542400
  number of nonzeros: 96845792 (27.339 per Row)
Starting CSR
runtime CSR: 1.59161 sec. performance: 1.21696e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 1.58117 sec. performance: 1.22499e+10 Flops/s, overhead: 0.797961%
```

```
ptfs214h@f0757:~/SpMVMS$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/parabolic_fem.mtx 8 1
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 525825x525825
  number of nonzeros: 3674625 (6.9883 per Row)
Starting CSR
runtime CSR: 0.0447869 sec. performance: 1.64094e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 0.0383871 sec. performance: 1.91451e+10 Flops/s, overhead: 0.0184747%
```

```
ptfs214h@f0757:~/SpMVMS$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/RM07R.mtx 8 1
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 381689x381689
  number of nonzeros: 37464962 (98.1557 per Row)
Starting CSR
runtime CSR: 0.543129 sec. performance: 1.3796e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 0.796595 sec. performance: 9.40628e+09 Flops/s, overhead: 32.453%
```

Sigma = 2

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/cage14.mtx 8 2

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 2

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/nlpkkt120.mtx 8 2

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/parabolic_fem.mtx 8 2

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/RM07R.mtx 8 2

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/cage14.mtx 8 2
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 1505785x1505785
  number of nonzeros: 27130349 (18.0174 per Row)
Starting CSR
runtime CSR: 0.490313 sec. performance: 1.10665e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-2
runtime Sell-8-2: 0.497055 sec. performance: 1.09164e+10 Flops/s, overhead: 2.73214%
```

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 2
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 1504002x1504002
  number of nonzeros: 110879972 (73.7233 per Row)
Starting CSR
runtime CSR: 1.65067 sec. performance: 1.34346e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-2
runtime Sell-8-2: 1.64814 sec. performance: 1.34552e+10 Flops/s, overhead: 0.274206%
```

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/nlpkkt120.mtx 8 2
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 3542400x3542400
  number of nonzeros: 96845792 (27.339 per Row)
Starting CSR
runtime CSR: 1.58459 sec. performance: 1.22235e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-2
runtime Sell-8-2: 1.58586 sec. performance: 1.22137e+10 Flops/s, overhead: 0.797961%
```

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/parabolic_fem.mtx 8 2
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 525825x525825
  number of nonzeros: 3674625 (6.9883 per Row)
Starting CSR
runtime CSR: 0.043165 sec. performance: 1.7026e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-2
runtime Sell-8-2: 0.038979 sec. performance: 1.88501e+10 Flops/s, overhead: 0.0184747%
```

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/RM07R.mtx 8 2
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 381689x381689
  number of nonzeros: 37464962 (98.1557 per Row)
Starting CSR
runtime CSR: 0.546106 sec. performance: 1.37208e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-2
runtime Sell-8-2: 0.793955 sec. performance: 9.43755e+09 Flops/s, overhead: 32.4541%
```

Sigma = 4

```
OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/cage14.mtx 8 4
```

```
OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 4
```

```
OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/nlpkkt120.mtx 8 4
```

```
OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/parabolic_fem.mtx 8 4
```

```
OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/RM07R.mtx 8 4
```

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/cage14.mtx 8 4
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 1505785x1505785
  number of nonzeros: 27130349 (18.0174 per Row)
Starting CSR
runtime CSR: 0.493846 sec. performance: 1.09874e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-4
runtime Sell-8-4: 0.497436 sec. performance: 1.09081e+10 Flops/s, overhead: 2.73214%
```

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 4
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 1504002x1504002
  number of nonzeros: 110879972 (73.7233 per Row)
Starting CSR
runtime CSR: 1.6526 sec. performance: 1.34188e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-4
runtime Sell-8-4: 1.64769 sec. performance: 1.34588e+10 Flops/s, overhead: 0.274264%
```

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/nlpkkt120.mtx 8 4
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 3542400x3542400
  number of nonzeros: 96845792 (27.339 per Row)
Starting CSR
runtime CSR: 1.59633 sec. performance: 1.21336e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-4
runtime Sell-8-4: 1.57522 sec. performance: 1.22962e+10 Flops/s, overhead: 0.797961%
```

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/parabolic_fem.mtx 8 4
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 525825x525825
  number of nonzeros: 3674625 (6.9883 per Row)
Starting CSR
runtime CSR: 0.043292 sec. performance: 1.6976e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-4
runtime Sell-8-4: 0.0401449 sec. performance: 1.83068e+10 Flops/s, overhead: 0.0184747%
```

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/RM07R.mtx 8 4
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 381689x381689
  number of nonzeros: 37464962 (98.1557 per Row)
Starting CSR
runtime CSR: 0.535929 sec. performance: 1.39813e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-4
runtime Sell-8-4: 0.795622 sec. performance: 9.41778e+09 Flops/s, overhead: 32.4541%
```

Sigma = 8

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/cage14.mtx 8 8

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 8

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/nlpkkt120.mtx 8 8

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/parabolic_fem.mtx 8 8

OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/RM07R.mtx 8 8

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/cage14.mtx 8 8
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 1505785x1505785
  number of nonzeros: 27130349 (18.0174 per Row)
Starting CSR
runtime CSR: 0.495789 sec. performance: 1.09443e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-8
runtime Sell-8-8: 0.499325 sec. performance: 1.08668e+10 Flops/s, overhead: 2.73214%
```

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 8
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 1504002x1504002
  number of nonzeros: 110879972 (73.7233 per Row)
Starting CSR
runtime CSR: 1.65653 sec. performance: 1.3387e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-8
runtime Sell-8-8: 1.64964 sec. performance: 1.34429e+10 Flops/s, overhead: 0.274307%
```

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/nlpkkt120.mtx 8 8
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 3542400x3542400
  number of nonzeros: 96845792 (27.339 per Row)
Starting CSR
runtime CSR: 1.59805 sec. performance: 1.21205e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-8
runtime Sell-8-8: 1.58015 sec. performance: 1.22578e+10 Flops/s, overhead: 0.797961%
```

```
ptfs214h@f0757:~/SpMVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark ~ptfs100h/Matrices/parabolic_fem.mtx 8 8
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 525825x525825
  number of nonzeros: 3674625 (6.9883 per Row)
Starting CSR
runtime CSR: 0.042624 sec. performance: 1.7242e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-8
runtime Sell-8-8: 0.0416989 sec. performance: 1.76246e+10 Flops/s, overhead: 0.0184747%
```

```
ptfs214h@f0757:~/SpHVM$ OMP_NUM_THREADS=18 OMP_PROC_BIND=close OMP_PLACES=cores OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 ./benchmark -ptfs100h/Matrices/RM07R.mtx 8 8
Running without Marker API. Activate Marker API with -m on commandline.
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 381689x381689
  number of nonzeros: 37464962 (98.1557 per Row)
Starting CSR
runtime CSR: 0.539225 sec. performance: 1.38959e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-8
runtime Sell-8-8: 0.799919 sec. performance: 9.36719e+09 Flops/s, overhead: 32.4541%
```

In the SELL-C- σ storage format, the parameter sigma (σ) represents the sorting scope, which determines the maximum number of non-zero elements considered for sorting within each chunk. The sorting scope impacts the trade-off between memory consumption and computational efficiency.

A larger sorting scope ($\sigma > 1$) allows for a wider range of non-zero elements to be considered for sorting within each chunk. This can potentially improve memory access patterns and exploit more parallelism in certain scenarios, resulting in performance gains. However, the benefits of a larger sigma depend on the characteristics of the sparse matrix and the target hardware architecture.

Determining whether a matrix would benefit from a larger sorting scope ($\sigma > 1$) involves analyzing its sparsity pattern and the memory access patterns during computation. The following considerations can help in assessing the potential performance gain:

1. **Sparsity Pattern:** If the matrix exhibits irregular or clustered non-zero patterns, a larger sorting scope can improve cache utilization and reduce memory access irregularities, resulting in performance gains.
2. **Chunk Size:** The size of the chunks relative to the matrix structure can influence the effectiveness of a larger sigma. If the chunk size is too small compared to the number of non-zero elements per row, increasing sigma may not provide significant benefits.
3. **Hardware Architecture:** The effectiveness of a larger sigma can vary depending on the specific hardware architecture being used. GPUs, for example, often have memory access patterns that can benefit from larger sorting scopes, especially when there is sufficient parallelism available.
4. **Computational Operations:** The type of operations being performed on the matrix also plays a role. Certain algorithms or computations may exhibit better performance with larger sigma, while others may not see significant gains.

Benchmarking and experimentation are typically required to evaluate the impact of different sorting scopes on a specific matrix and computation. By measuring execution times and performance metrics for varying sigma values, one can determine the optimal sorting scope that achieves the best trade-off between memory usage and computational efficiency for a given scenario.

```
OMP_NUM_THREADS=18 OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/cage14.mtx 8 1
```

```
OMP_NUM_THREADS=18 OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 1
```

```
OMP_NUM_THREADS=18 OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/parabolic_fem.mtx 8 1
```

OMP_NUM_THREADS=18 OMP_SCHEDULE=guided srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/cage14.mtx 8 1

OMP_NUM_THREADS=18 OMP_SCHEDULE=guided srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 1

OMP_NUM_THREADS=18 OMP_SCHEDULE=guided srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/parabolic_fem.mtx 8 1

cage14 performance stats CSR:

```
ptfs214h@f0907:~/SpMVM$ OMP_NUM_THREADS=18 OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/cage14.mtx 8 1
-----
CPU name:      Intel(R) Xeon(R) Platinum 8360Y CPU @ 2.40GHz
CPU type:      Intel Icelake SP processor
CPU clock:     2.39 GHz
-----
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 1505785x1505785
  number of nonzeros: 27130349 (18.0174 per Row)
Starting CSR
runtime CSR: 0.486419 sec. performance: 1.11551e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 0.50748 sec. performance: 1.06922e+10 Flops/s, overhead: 2.73208%
```

```
ptfs214h@f0907:~/SpMVM$ OMP_NUM_THREADS=18 OMP_SCHEDULE=guided srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/parabolic_fem.mtx 8 1
-----
CPU name:      Intel(R) Xeon(R) Platinum 8360Y CPU @ 2.40GHz
CPU type:      Intel Icelake SP processor
CPU clock:     2.39 GHz
-----
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (guided, 1)
  matrix size: 525825x525825
  number of nonzeros: 3674625 (6.9883 per Row)
Starting CSR
runtime CSR: 0.0475569 sec. performance: 1.54536e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 0.0409398 sec. performance: 1.79514e+10 Flops/s, overhead: 0.0184747%
```

static

Event	Counter	Sum	Min	Max	Avg
INSTR_RETIRED_ANY STAT	FIXC0	14652608700	768973400	984390500	8.140338e+08
CPU_CLK_UNHALTED_CORE STAT	FIXC1	20788023000	1143078000	1159145000	1.154890e+09
CPU_CLK_UNHALTED_REF STAT	FIXC2	20788012000	1143073000	1159145000	1.154890e+09
TOPDOWN_SLOTS STAT	FIXC3	103940127000	5715390000	5795726000	5774451500
PWR_PKG_ENERGY STAT	PWR0	77.8534	0	77.8534	4.3252
PWR_DRAM_ENERGY STAT	PWR3	7.0937	0	7.0937	0.3941
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE STAT	PMC0	148888900	7546300	8365500	8.271606e+06
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE STAT	PMC1	1354113740	63647230	81745630	7.522854e+07
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE STAT	PMC2	1241322500	38345900	95318600	6.896236e+07
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE STAT	PMC3	0	0	0	0
CAS_COUNT_RD STAT	MBOX0C0	142176800	0	142176800	7.898711e+06
CAS_COUNT_WR STAT	MBOX0C0	4413610	0	4413610	245200.5556
CAS_COUNT_RD STAT	MBOX1C0	142168000	0	142168000	7.898222e+06
CAS_COUNT_WR STAT	MBOX1C1	4409991	0	4409991	244999.5000
CAS_COUNT_RD STAT	MBOX2C0	142168500	0	142168500	7898250
CAS_COUNT_WR STAT	MBOX2C1	4402878	0	4402878	244604.3333
CAS_COUNT_RD STAT	MBOX3C0	142228200	0	142228200	7.901567e+06
CAS_COUNT_WR STAT	MBOX3C1	4411454	0	4411454	245080.7778
CAS_COUNT_RD STAT	MBOX4C0	1920	0	1920	106.6667
CAS_COUNT_WR STAT	MBOX4C1	1846	0	1846	102.5556
CAS_COUNT_RD STAT	MBOX5C0	3836	0	3836	213.1111
CAS_COUNT_WR STAT	MBOX5C1	2169	0	2169	120.5000
CAS_COUNT_RD STAT	MBOX6C0	2174	0	2174	120.7778
CAS_COUNT_WR STAT	MBOX6C1	2147	0	2147	119.2778
CAS_COUNT_RD STAT	MBOX7C0	3521	0	3521	195.6111
CAS_COUNT_WR STAT	MBOX7C1	1990	0	1990	110.5556

guided

Event	Counter	Sum	Min	Max	Avg
INSTR_RETIRED_ANY STAT	FIXC0	3283974200	177161200	186786000	1.824430e+08
CPU_CLK_UNHALTED_CORE STAT	FIXC1	2002427600	110697600	111320900	1.112460e+08
CPU_CLK_UNHALTED_REF STAT	FIXC2	2002427700	110698000	111320700	1.112460e+08
TOPDOWN_SLOTS STAT	FIXC3	10012137200	553487800	556604700	5.562298e+08
PWR_PKG_ENERGY STAT	PWR0	8.1867	0	8.1867	0.4548
PWR_DRAM_ENERGY STAT	PWR3	0.6564	0	0.6564	0.0365
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE STAT	PMC0	0	0	0	0
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE STAT	PMC1	734971230	39766160	41801680	40831735
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE STAT	PMC2	0	0	0	0
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE STAT	PMC3	0	0	0	0
CAS_COUNT_RD STAT	MBOX0C0	10682470	0	10682470	593470.5556
CAS_COUNT_WR STAT	MBOX0C1	567062	0	567062	31503.4444
CAS_COUNT_RD STAT	MBOX1C0	10660650	0	10660650	592258.3333
CAS_COUNT_WR STAT	MBOX1C1	563490	0	563490	31305
CAS_COUNT_RD STAT	MBOX2C0	10682910	0	10682910	593495
CAS_COUNT_WR STAT	MBOX2C1	564895	0	564895	31383.0556
CAS_COUNT_RD STAT	MBOX3C0	10695930	0	10695930	594218.3333
CAS_COUNT_WR STAT	MBOX3C1	562929	0	562929	31273.8333
CAS_COUNT_RD STAT	MBOX4C0	1654	0	1654	91.8889
CAS_COUNT_WR STAT	MBOX4C1	192	0	192	10.6667
CAS_COUNT_RD STAT	MBOX5C0	2376	0	2376	132
CAS_COUNT_WR STAT	MBOX5C1	253	0	253	14.0556
CAS_COUNT_RD STAT	MBOX6C0	2429	0	2429	134.9444
CAS_COUNT_WR STAT	MBOX6C1	255	0	255	14.1667
CAS_COUNT_RD STAT	MBOX7C0	2864	0	2864	159.1111
CAS_COUNT_WR STAT	MBOX7C1	222	0	222	12.3333

SELL -C-SIGMA

static

Event	Counter	Sum	Min	Max	Avg
INSTR_RETIRED_ANY STAT	FIXC0	7872003400	338185800	656532000	4.373335e+08
CPU_CLK_UNHALTED_CORE STAT	FIXC1	21657732000	1191238000	1209837000	1.203207e+09
CPU_CLK_UNHALTED_REF STAT	FIXC2	21657729000	1191231000	1209838000	1.203207e+09
TOPDOWN_SLOTS STAT	FIXC3	108288670000	5956191000	6049186000	6.016037e+09
PWR_PKG_ENERGY STAT	PWR0	79.3816	0	79.3816	4.4101
PWR_DRAM_ENERGY STAT	PWR3	7.2563	0	7.2563	0.4031
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE STAT	PMC0	0	0	0	0
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE STAT	PMC1	576	32	32	32
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE STAT	PMC2	1394619600	43297200	106589600	7.747887e+07
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE STAT	PMC3	0	0	0	0
CAS_COUNT_RD STAT	MBOX0C0	144343300	0	144343300	8.019072e+06
CAS_COUNT_WR STAT	MBOX0C1	4420207	0	4420207	245567.0556
CAS_COUNT_RD STAT	MBOX1C0	144294600	0	144294600	8.016367e+06
CAS_COUNT_WR STAT	MBOX1C1	4406251	0	4406251	244791.7222
CAS_COUNT_RD STAT	MBOX2C0	144334600	0	144334600	8.018589e+06
CAS_COUNT_WR STAT	MBOX2C1	4405590	0	4405590	244755
CAS_COUNT_RD STAT	MBOX3C0	144352400	0	144352400	8.019578e+06
CAS_COUNT_WR STAT	MBOX3C1	4411319	0	4411319	245073.2778
CAS_COUNT_RD STAT	MBOX4C0	2204	0	2204	122.4444
CAS_COUNT_WR STAT	MBOX4C1	2147	0	2147	119.2778
CAS_COUNT_RD STAT	MBOX5C0	4127	0	4127	229.2778
CAS_COUNT_WR STAT	MBOX5C1	2814	0	2814	156.3333
CAS_COUNT_RD STAT	MBOX6C0	2686	0	2686	149.2222
CAS_COUNT_WR STAT	MBOX6C1	2734	0	2734	151.8889
CAS_COUNT_RD STAT	MBOX7C0	3806	0	3806	211.4444
CAS_COUNT_WR STAT	MBOX7C1	2540	0	2540	141.1111

guided

Event	Counter	Sum	Min	Max	Avg
INSTR_RETIRED_ANY STAT	FIXC0	894355390	47024280	50565710	4.968641e+07
CPU_CLK_UNHALTED_CORE STAT	FIXC1	1723534550	95216160	95817280	9.575192e+07
CPU_CLK_UNHALTED_REF STAT	FIXC2	1723532350	95216740	95817410	9.575180e+07
TOPDOWN_SLOTS STAT	FIXC3	8617672700	476080800	479086400	4.787596e+08
PWR_PKG_ENERGY STAT	PWR0	6.9165	0	6.9165	0.3842
PWR_DRAM_ENERGY STAT	PWR3	0.5772	0	0.5772	0.0321
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE STAT	PMC0	0	0	0	0
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE STAT	PMC1	33548	1754	1929	1863.7778
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE STAT	PMC2	183765190	9799140	10383680	1.020918e+07
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE STAT	PMC3	0	0	0	0
CAS_COUNT_RD STAT	MBOX0C0	9761773	0	9761773	542320.7222
CAS_COUNT_WR STAT	MBOX0C1	558547	0	558547	31030.3889
CAS_COUNT_RD STAT	MBOX1C0	9790064	0	9790064	543892.4444
CAS_COUNT_WR STAT	MBOX1C1	553238	0	553238	30735.4444
CAS_COUNT_RD STAT	MBOX2C0	9758990	0	9758990	542166.1111
CAS_COUNT_WR STAT	MBOX2C1	554549	0	554549	30808.2778
CAS_COUNT_RD STAT	MBOX3C0	9797328	0	9797328	544296
CAS_COUNT_WR STAT	MBOX3C1	554663	0	554663	30814.6111
CAS_COUNT_RD STAT	MBOX4C0	1526	0	1526	84.7778
CAS_COUNT_WR STAT	MBOX4C1	166	0	166	9.2222
CAS_COUNT_RD STAT	MBOX5C0	2194	0	2194	121.8889
CAS_COUNT_WR STAT	MBOX5C1	201	0	201	11.1667
CAS_COUNT_RD STAT	MBOX6C0	2213	0	2213	122.9444
CAS_COUNT_WR STAT	MBOX6C1	198	0	198	11
CAS_COUNT_RD STAT	MBOX7C0	2711	0	2711	150.6111
CAS_COUNT_WR STAT	MBOX7C1	179	0	179	9.9444

ML_Geer performance stats CSR:

```
ptfs214h@f0907:~/SpMVM$ OMP_NUM_THREADS=18 OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark -ptfs100h/Matrices/ML_Geer.mtx 8 1
-----
CPU name:      Intel(R) Xeon(R) Platinum 8360Y CPU @ 2.40GHz
CPU type:      Intel Icelake SP processor
CPU clock:     2.39 GHz
-----
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 1504002x1504002
  number of nonzeros: 110879972 (73.7233 per Row)
Starting CSR
runtime CSR: 1.66601 sec. performance: 1.33108e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 1.65243 sec. performance: 1.34203e+10 Flops/s, overhead: 0.274206%
-----
ptfs214h@f0907:~/SpMVM$ OMP_NUM_THREADS=18 OMP_SCHEDULE=guided srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark -ptfs100h/Matrices/ML_Geer.mtx 8 1
-----
CPU name:      Intel(R) Xeon(R) Platinum 8360Y CPU @ 2.40GHz
CPU type:      Intel Icelake SP processor
CPU clock:     2.39 GHz
-----
Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (guided, 1)
  matrix size: 1504002x1504002
  number of nonzeros: 110879972 (73.7233 per Row)
Starting CSR
runtime CSR: 1.65537 sec. performance: 1.33964e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 1.65005 sec. performance: 1.34396e+10 Flops/s, overhead: 0.274206%
```


Event	Counter	Sum	Min	Max	Avg
INSTR_RETIRED_ANY STAT	FIXC0	26344194000	1399218000	1564285000	1.463566e+09
CPU_CLK_UNHALTED_CORE STAT	FIXC1	71444764000	3962324000	3972275000	3.969154e+09
CPU_CLK_UNHALTED_REF STAT	FIXC2	71444762000	3962322000	3972273000	3.969153e+09
TOPDOWN_SLOTS STAT	FIXC3	357223800000	19811620000	19861380000	1.984577e+10
PWR_PKG_ENERGY STAT	PWR0	260.5336	0	260.5336	14.4741
PWR_DRAM_ENERGY STAT	PWR3	24.6250	0	24.6250	1.3681
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE STAT	PMC0	150400200	8355500	8355600	8.355567e+06
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE STAT	PMC1	908779510	50335230	51227430	5.048775e+07
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE STAT	PMC2	5542404200	305279900	308451600	3.079113e+08
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE STAT	PMC3	0	0	0	0
CAS_COUNT_RD STAT	MBOX0C0	538062100	0	538062100	2.989234e+07
CAS_COUNT_WR STAT	MBOX0C1	5066726	0	5066726	281484.7778
CAS_COUNT_RD STAT	MBOX1C0	537921900	0	537921900	29884550
CAS_COUNT_WR STAT	MBOX1C1	4981288	0	4981288	276738.2222
CAS_COUNT_RD STAT	MBOX2C0	538067800	0	538067800	2.989266e+07
CAS_COUNT_WR STAT	MBOX2C1	5017644	0	5017644	278758
CAS_COUNT_RD STAT	MBOX3C0	538147600	0	538147600	2.989709e+07
CAS_COUNT_WR STAT	MBOX3C1	4991450	0	4991450	277302.7778
CAS_COUNT_RD STAT	MBOX4C0	7040	0	7040	391.1111
CAS_COUNT_WR STAT	MBOX4C1	6315	0	6315	350.8333
CAS_COUNT_RD STAT	MBOX5C0	24923	0	24923	1384.6111
CAS_COUNT_WR STAT	MBOX5C1	7580	0	7580	421.1111
CAS_COUNT_RD STAT	MBOX6C0	8269	0	8269	459.3889
CAS_COUNT_WR STAT	MBOX6C1	7569	0	7569	420.5000
CAS_COUNT_RD STAT	MBOX7C0	19217	0	19217	1067.6111
CAS_COUNT_WR STAT	MBOX7C1	7094	0	7094	394.1111

SELL C SIGMA

Event	Counter	Sum	Min	Max	Avg
INSTR_RETIRED_ANY STAT	FIXC0	16085229600	861142200	1008815000	8.936239e+08
CPU_CLK_UNHALTED_CORE STAT	FIXC1	70925053000	3934659000	3942284000	3.940281e+09
CPU_CLK_UNHALTED_REF STAT	FIXC2	70925041000	3934654000	3942286000	3.940280e+09
TOPDOWN_SLOTS STAT	FIXC3	354625260000	19673290000	19711420000	1.970140e+10
PWR_PKG_ENERGY STAT	PWR0	255.6069	0	255.6069	14.2004
PWR_DRAM_ENERGY STAT	PWR3	24.6311	0	24.6311	1.3684
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE STAT	PMC0	0	0	0	0
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE STAT	PMC1	577	32	33	32.0556
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE STAT	PMC2	5559242400	307976800	309031200	308846800
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE STAT	PMC3	0	0	0	0
CAS_COUNT_RD STAT	MBOX0C0	537356200	0	537356200	2.985312e+07
CAS_COUNT_WR STAT	MBOX0C1	4985426	0	4985426	276968.1111
CAS_COUNT_RD STAT	MBOX1C0	537321700	0	537321700	2.985121e+07
CAS_COUNT_WR STAT	MBOX1C1	4956800	0	4956800	275377.7778
CAS_COUNT_RD STAT	MBOX2C0	537433600	0	537433600	2.985742e+07
CAS_COUNT_WR STAT	MBOX2C1	4956013	0	4956013	275334.0556
CAS_COUNT_RD STAT	MBOX3C0	537464000	0	537464000	2.985911e+07
CAS_COUNT_WR STAT	MBOX3C1	4959338	0	4959338	275518.7778
CAS_COUNT_RD STAT	MBOX4C0	8009	0	8009	444.9444
CAS_COUNT_WR STAT	MBOX4C1	8134	0	8134	451.8889
CAS_COUNT_RD STAT	MBOX5C0	26540	0	26540	1474.4444
CAS_COUNT_WR STAT	MBOX5C1	9993	0	9993	555.1667
CAS_COUNT_RD STAT	MBOX6C0	9285	0	9285	515.8333
CAS_COUNT_WR STAT	MBOX6C1	9692	0	9692	538.4444
CAS_COUNT_RD STAT	MBOX7C0	18807	0	18807	1044.8333
CAS_COUNT_WR STAT	MBOX7C1	8850	0	8850	491.6667

parabolic_fem performance stats CSR:

```
ptfs214h@f0907:~/SpMVM$ OMP_NUM_THREADS=18 OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark -ptfs100h/Matrices/parabolic_fem.mtx 8 1
-----
CPU name:      Intel(R) Xeon(R) Platinum 8360Y CPU @ 2.40GHz
CPU type:      Intel Icelake SP processor
CPU clock:     2.39 GHz
-----

Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (static, 0)
  matrix size: 525825x525825
  number of nonzeros: 3674625 (6.9883 per Row)

Starting CSR
runtime CSR: 0.04461 sec. performance: 1.64744e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 0.0411899 sec. performance: 1.78424e+10 Flops/s, overhead: 0.0184747%
-----

ptfs214h@f0907:~/SpMVM$ OMP_NUM_THREADS=18 OMP_SCHEDULE=guided srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0-17 -g MEM_DP -m ./benchmark -ptfs100h/Matrices/parabolic_fem.mtx 8 1
-----
CPU name:      Intel(R) Xeon(R) Platinum 8360Y CPU @ 2.40GHz
CPU type:      Intel Icelake SP processor
CPU clock:     2.39 GHz
-----

Matrix Format Benchmark:
  revisions: 100
  number of threads: 18
  schedule: (guided, 1)
  matrix size: 525825x525825
  number of nonzeros: 3674625 (6.9883 per Row)

Starting CSR
runtime CSR: 0.045738 sec. performance: 1.60682e+10 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 0.0481549 sec. performance: 1.83822e+10 Flops/s, overhead: 0.0184747%
-----
```

Event	Counter	Sum	Min	Max	Avg
INSTR_RETIRED_ANY STAT	FIXC0	3432990300	180403100	198455300	1.907217e+08
CPU_CLK_UNHALTED_CORE STAT	FIXC1	1873554900	103757800	104481500	1.040864e+08
CPU_CLK_UNHALTED_REF STAT	FIXC2	1873557200	103757600	104482100	1.040865e+08
TOPDOWN_SLOTS STAT	FIXC3	9367774500	518789000	522407400	5.204319e+08
PWR_PKG_ENERGY STAT	PWR0	7.8757	0	7.8757	0.4375
PWR_DRAM_ENERGY STAT	PWR3	0.6242	0	0.6242	0.0347
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE STAT	PMC0	0	0	0	0
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE STAT	PMC1	734925540	40282430	40898230	4.082920e+07
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE STAT	PMC2	0	0	0	0
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE STAT	PMC3	0	0	0	0
CAS_COUNT_RD STAT	MBOX0C0	10477570	0	10477570	582087.2222
CAS_COUNT_WR STAT	MBOX0C1	690174	0	690174	38343
CAS_COUNT_RD STAT	MBOX1C0	10460440	0	10460440	581135.5556
CAS_COUNT_WR STAT	MBOX1C1	684231	0	684231	38012.8333
CAS_COUNT_RD STAT	MBOX2C0	10471910	0	10471910	581772.7778
CAS_COUNT_WR STAT	MBOX2C1	690941	0	690941	38385.6111
CAS_COUNT_RD STAT	MBOX3C0	10492800	0	10492800	582933.3333
CAS_COUNT_WR STAT	MBOX3C1	683285	0	683285	37960.2778
CAS_COUNT_RD STAT	MBOX4C0	655	0	655	36.3889
CAS_COUNT_WR STAT	MBOX4C1	189	0	189	10.5000
CAS_COUNT_RD STAT	MBOX5C0	1141	0	1141	63.3889
CAS_COUNT_WR STAT	MBOX5C1	274	0	274	15.2222
CAS_COUNT_RD STAT	MBOX6C0	827	0	827	45.9444
CAS_COUNT_WR STAT	MBOX6C1	283	0	283	15.7222
CAS_COUNT_RD STAT	MBOX7C0	2626	0	2626	145.8889
CAS_COUNT_WR STAT	MBOX7C1	246	0	246	13.6667

SELL C SIGMA

Event	Counter	Sum	Min	Max	Avg
INSTR_RETIRED_ANY STAT	FIXC0	1115902910	48824600	71573240	6.199461e+07
CPU_CLK_UNHALTED_CORE STAT	FIXC1	1726297500	95523340	96469670	9.590542e+07
CPU_CLK_UNHALTED_REF STAT	FIXC2	1726297550	95522980	96469540	9.590542e+07
TOPDOWN_SLOTS STAT	FIXC3	8631487600	477616700	482348400	4.795271e+08
PWR_PKG_ENERGY STAT	PWR0	7.1080	0	7.1080	0.3949
PWR_DRAM_ENERGY STAT	PWR3	0.5758	0	0.5758	0.0320
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE STAT	PMC0	0	0	0	0
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE STAT	PMC1	576	32	32	32
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE STAT	PMC2	183765200	10074800	10225600	1.020918e+07
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE STAT	PMC3	0	0	0	0
CAS_COUNT_RD STAT	MBOX0C0	9454714	0	9454714	525261.8889
CAS_COUNT_WR STAT	MBOX0C1	581394	0	581394	32299.6667
CAS_COUNT_RD STAT	MBOX1C0	9429981	0	9429981	523887.8333
CAS_COUNT_WR STAT	MBOX1C1	575279	0	575279	31959.9444
CAS_COUNT_RD STAT	MBOX2C0	9445495	0	9445495	524749.7222
CAS_COUNT_WR STAT	MBOX2C1	579578	0	579578	32198.7778
CAS_COUNT_RD STAT	MBOX3C0	9446277	0	9446277	524793.1667
CAS_COUNT_WR STAT	MBOX3C1	576609	0	576609	32033.8333
CAS_COUNT_RD STAT	MBOX4C0	625	0	625	34.7222
CAS_COUNT_WR STAT	MBOX4C1	161	0	161	8.9444
CAS_COUNT_RD STAT	MBOX5C0	1000	0	1000	55.5556
CAS_COUNT_WR STAT	MBOX5C1	196	0	196	10.8889
CAS_COUNT_RD STAT	MBOX6C0	686	0	686	38.1111
CAS_COUNT_WR STAT	MBOX6C1	189	0	189	10.5000
CAS_COUNT_RD STAT	MBOX7C0	2576	0	2576	143.1111
CAS_COUNT_WR STAT	MBOX7C1	179	0	179	9.9444

Final part

With simd

```
ptfs214h@f0907:~/SpMVM$ OMP_SCHEDULE=static srtn --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0 -g MEM_DP -m ./benchmark -ptfs100h/Matrices/ML_Geer.mtx 8 1
CPU name:      Intel(R) Xeon(R) Platinum 8360Y CPU @ 2.40GHz
CPU type:      Intel Icelake SP processor
CPU clock:     2.39 GHz
```

```
-----
Matrix Format Benchmark:
  revisions: 100
  number of threads: 1
  schedule: (static, 0)
  matrix size: 1504002x1504002
  number of nonzeros: 110879972 (73.7233 per Row)
Starting CSR
runtime CSR: 8.87966 sec. performance: 2.49739e+09 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 8.75242 sec. performance: 2.5337e+09 Flops/s, overhead: 0.274206%
-----
```

Region SpMV_CSR, Group 1: MEM_DP

Region Info	HWThread 0
RDTS Runtime [s]	8.878868
call count	1

Event	Counter	HWThread 0
INSTR_RETIRED_ANY	FIXC0	25063340000
CPU_CLK_UNHALTED_CORE	FIXC1	21206410000
CPU_CLK_UNHALTED_REF	FIXC2	21206410000
TOPDOWN_SLOTS	FIXC3	106032000000
PWR_PKG_ENERGY	PWR0	950.6282
PWR_DRAM_ENERGY	PWR3	79.0473
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE	PMC0	150400200
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE	PMC1	908778800
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE	PMC2	5542404000
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE	PMC3	0
CAS_COUNT_RD	MBOX0C0	532534500
CAS_COUNT_WR	MBOX0C1	4397069
CAS_COUNT_RD	MBOX1C0	532495100
CAS_COUNT_WR	MBOX1C1	4376428
CAS_COUNT_RD	MBOX2C0	532493600
CAS_COUNT_WR	MBOX2C1	4382652
CAS_COUNT_RD	MBOX3C0	532560900
CAS_COUNT_WR	MBOX3C1	4382636
CAS_COUNT_RD	MBOX4C0	34730
CAS_COUNT_WR	MBOX4C1	33427

Region SpMV_SELL-C-SIGMA, Group 1: MEM_DP

Region Info		HWThread 0
RDTSC Runtime [s]	8.751627	
call count	1	

Event	Counter	HWThread 0
INSTR_RETIRED_ANY	FIXC0	15458770000
CPU_CLK_UNHALTED_CORE	FIXC1	20899520000
CPU_CLK_UNHALTED_REF	FIXC2	20899520000
TOPDOWN_SLOTS	FIXC3	104497600000
PWR_PKG_ENERGY	PWR0	936.9580
PWR_DRAM_ENERGY	PWR3	79.1300
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE	PMC0	0
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE	PMC1	32
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE	PMC2	5559242000
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE	PMC3	0
CAS_COUNT_RD	MBOX0C0	534304700
CAS_COUNT_WR	MBOX0C1	4474640
CAS_COUNT_RD	MBOX1C0	534298500
CAS_COUNT_WR	MBOX1C1	4456598
CAS_COUNT_RD	MBOX2C0	534321500
CAS_COUNT_WR	MBOX2C1	4463121
CAS_COUNT_RD	MBOX3C0	534340400
CAS_COUNT_WR	MBOX3C1	4460501
CAS_COUNT_RD	MBOX4C0	34112
CAS_COUNT_WR	MBOX4C1	33124
CAS_COUNT_RD	MBOX5C0	36597
CAS_COUNT_WR	MBOX5C1	37746
CAS_COUNT_RD	MBOX6C0	38160
CAS_COUNT_WR	MBOX6C1	38242
CAS_COUNT_RD	MBOX7C0	35440
CAS_COUNT_WR	MBOX7C1	35221

Metric	HWThread 0
Runtime (RDTSC) [s]	8.7516
Runtime unhalsted [s]	8.7287
Clock [MHz]	2394.3480
CPI	1.3520
Energy [J]	936.9580
Power [W]	107.0610
Energy DRAM [J]	79.1300

Without simd

```
ptfs214h@f0907:~/novect$ OMP_SCHEDULE=static srun --cpu-freq=2400000-2400000 likwid-perfctr -C S0:0 -g MEM_DP -m ./benchmark ~ptfs100h/Matrices/ML_Geer.mtx 8 1
-----
CPU name:      Intel(R) Xeon(R) Platinum 8360Y CPU @ 2.40GHz
CPU type:      Intel Icelake SP processor
CPU clock:      2.39 GHz
-----
Matrix Format Benchmark:
  revisions: 100
  number of threads: 1
  schedule: (static, 0)
  matrix size: 1504002x1504002
  number of nonzeros: 110879972 (73.7233 per Row)
Starting CSR
runtime CSR: 9.46296 sec. performance: 2.34345e+09 Flops/sec
Doing SELL-C-sigma...
Ctor matrix...
Ctor matrix done...
Starting Sell-8-1
runtime Sell-8-1: 10.6965 sec. performance: 2.0732e+09 Flops/s, overhead: 0.274206%
-----
Region SpMV_CSR, Group 1: MEM_DP
+-----+-----+
| Region Info | HWThread 0 |
+-----+-----+
| RDTSC Runtime [s] | 9.462174 |
| call count | 1 |
+-----+-----+

+-----+-----+-----+
| Event | Counter | HWThread 0 |
+-----+-----+-----+
| INSTR_RETIRED_ANY | FIXC0 | 5395857000 |
| CPU_CLK_UNHALTED_CORE | FIXC1 | 22599140000 |
| CPU_CLK_UNHALTED_REF | FIXC2 | 22599130000 |
| TOPDOWN_SLOTS | FIXC3 | 11299570000 |
| PWR_PKG_ENERGY | PWR0 | 1012.7440 |
| PWR_DRAM_ENERGY | PWR3 | 81.9298 |
| FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE | PMC0 | 2697597000 |
| FP_ARITH_INST_RETIRED_SCALAR_DOUBLE | PMC1 | 5990411000 |
| FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE | PMC2 | 2697597000 |
| FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE | PMC3 | 0 |
| CAS_COUNT_RD | MBOX0C0 | 533113800 |
| CAS_COUNT_WR | MBOX0C1 | 4467277 |
| CAS_COUNT_RD | MBOX1C0 | 533072700 |
| CAS_COUNT_WR | MBOX1C1 | 4448077 |
```

Region SpMV_SELL-C-SIGMA, Group 1: MEM_DP

Region Info	HWThread 0
RDTSC Runtime [s]	10.695760
call count	1

Event	Counter	HWThread 0
INSTR_RETIRED_ANY	FIXC0	42181930000
CPU_CLK_UNHALTED_CORE	FIXC1	25547090000
CPU_CLK_UNHALTED_REF	FIXC2	25547090000
TOPDOWN_SLOTS	FIXC3	127735400000
PWR_PKG_ENERGY	PWR0	1134.2340
PWR_DRAM_ENERGY	PWR3	92.9865
FP_ARITH_INST_RETIRED_128B_PACKED_DOUBLE	PMC0	0
FP_ARITH_INST_RETIRED_SCALAR_DOUBLE	PMC1	21631740000
FP_ARITH_INST_RETIRED_256B_PACKED_DOUBLE	PMC2	151306400
FP_ARITH_INST_RETIRED_512B_PACKED_DOUBLE	PMC3	0
CAS_COUNT_RD	MBOX0C0	532027900
CAS_COUNT_WR	MBOX0C1	4448764
CAS_COUNT_RD	MBOX1C0	532041900
CAS_COUNT_WR	MBOX1C1	4434505
CAS_COUNT_RD	MBOX2C0	532093400
CAS_COUNT_WR	MBOX2C1	4432846
CAS_COUNT_RD	MBOX3C0	532146000
CAS_COUNT_WR	MBOX3C1	4437882
CAS_COUNT_RD	MBOX4C0	42447
CAS_COUNT_WR	MBOX4C1	41253
CAS_COUNT_RD	MBOX5C0	45681
CAS_COUNT_WR	MBOX5C1	47149
CAS_COUNT_RD	MBOX6C0	48230
CAS_COUNT_WR	MBOX6C1	48237
CAS_COUNT_RD	MBOX7C0	44118
CAS_COUNT_WR	MBOX7C1	43482

Metric	HWThread 0
Runtime (RD TSC) [s]	10.6958
Runtime unhal ted [s]	10.6698
Clock [MHz]	2394.3277
CPI	0.6056